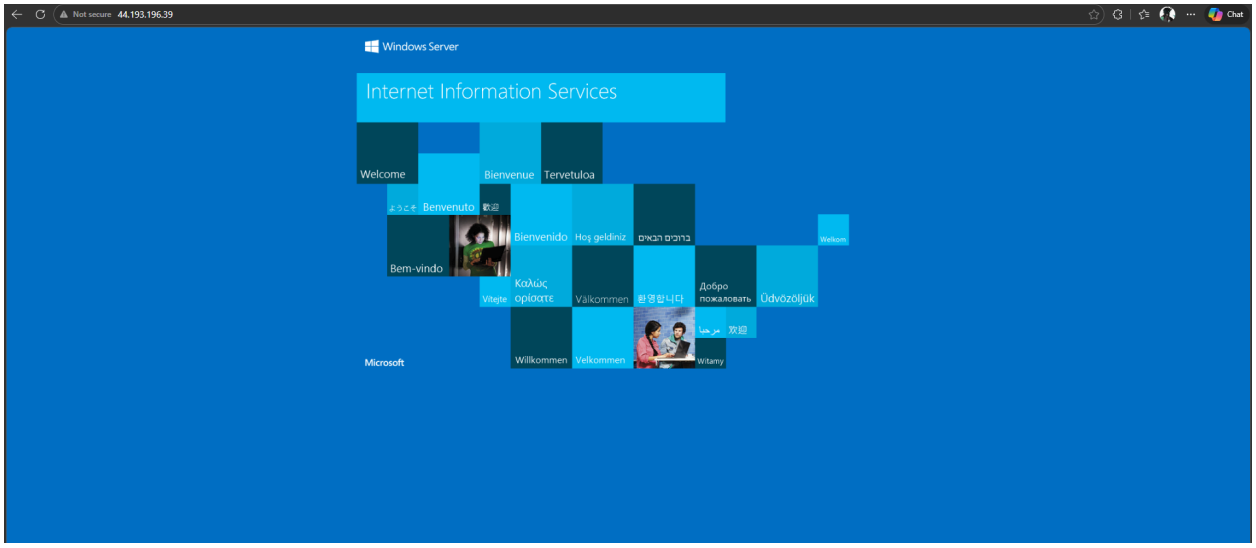
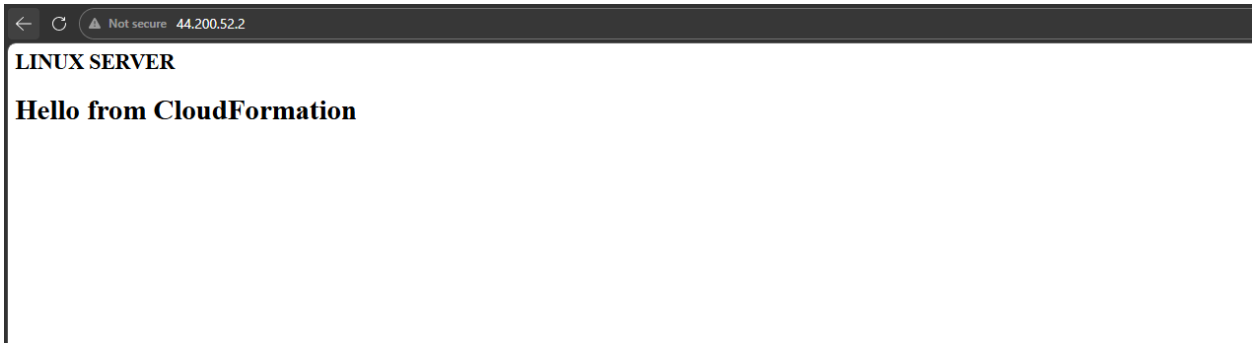
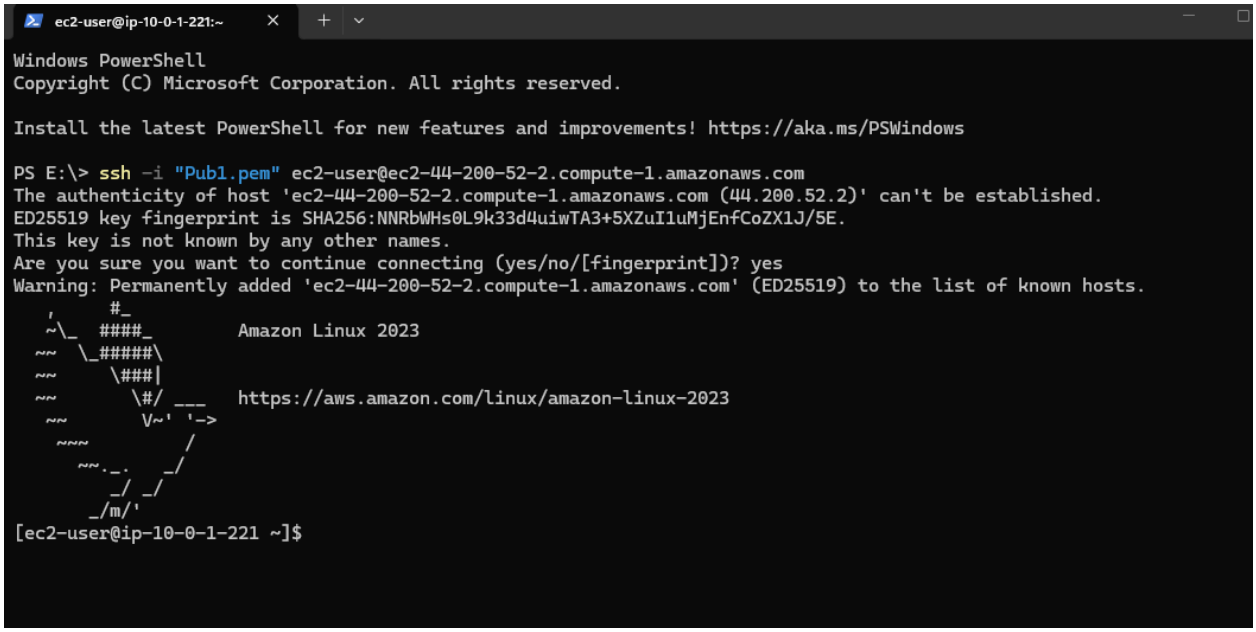
Os type launch using CFT

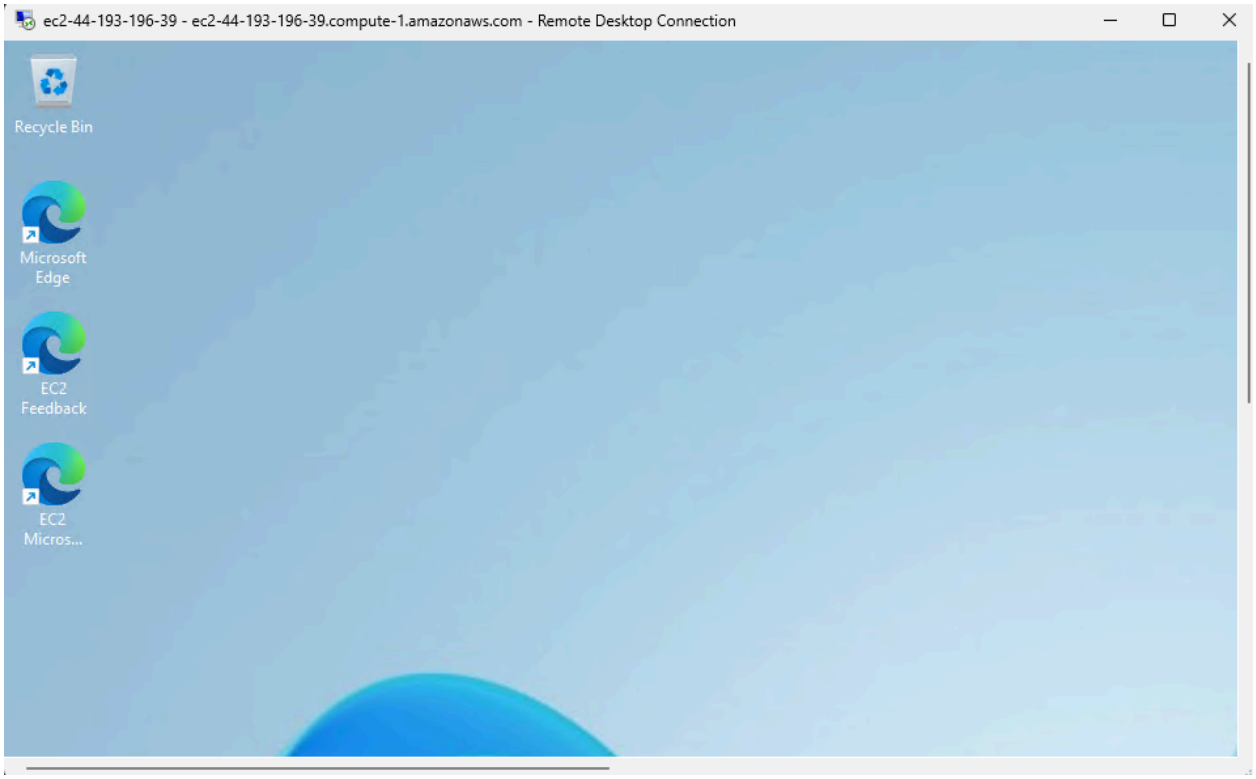
IIS web server



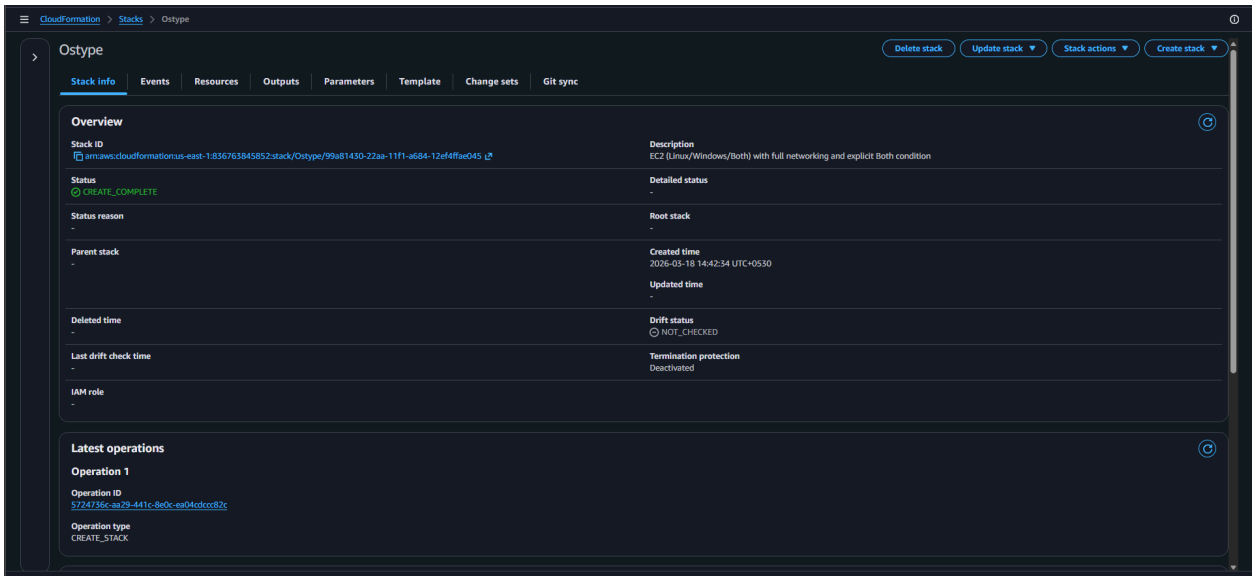
Apache server



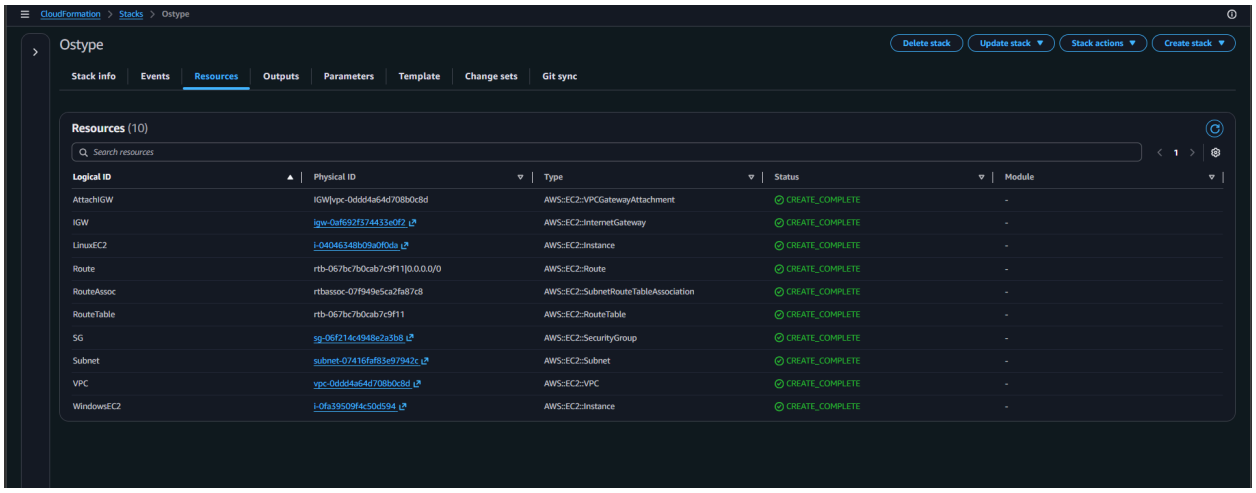

ec2-user@ip-10-0-1-221:~
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
PS E:\> ssh -i "Pub1.pem" ec2-user@ec2-44-200-52-2.compute-1.amazonaws.com
The authenticity of host 'ec2-44-200-52-2.compute-1.amazonaws.com (44.200.52.2)' can't be established.
ED25519 key fingerprint is SHA256:NNRbWHs0L9k33d4uiwTA3+5XZuI1uMjEnfCoZX1J/5E.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-44-200-52-2.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Amazon Linux 2023
https://aws.amazon.com/linux/amazon-linux-2023
[ec2-user@ip-10-0-1-221 ~]$


ec2-44-193-196-39 - ec2-44-193-196-39.compute-1.amazonaws.com - Remote Desktop Connection
Recycle Bin
Microsoft Edge
EC2 Feedback
EC2 Micros...

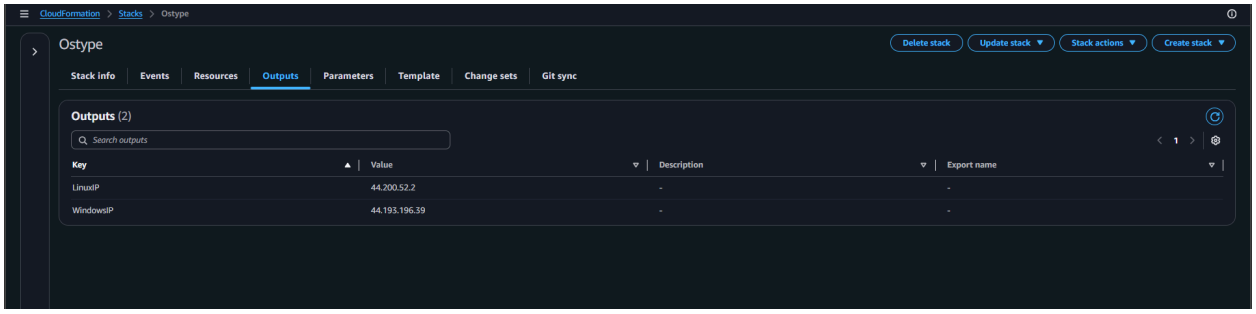
Stack info



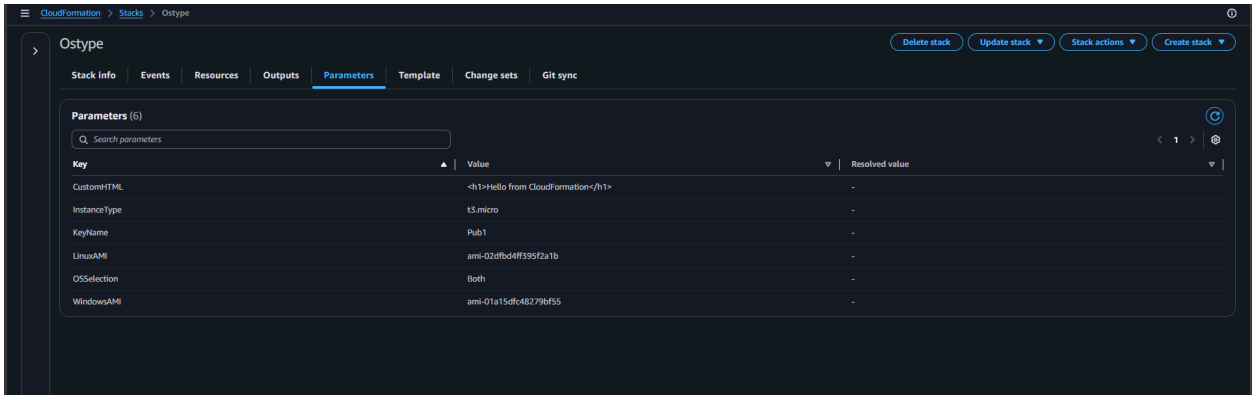
Resources



Output



Parameters



Template

```
AWSTemplateFormatVersion: '2010-09-09'
Description: EC2 (Linux/Windows/Both) with full networking and explicit Both condition

Parameters:
  OSSelection:
    Type: String
    Default: Linux
    AllowedValues:
      - Linux
      - Windows
```

```
      - Both

  InstanceType:
    Type: String
    Default: t3.micro

  KeyName:
    Type: AWS::EC2::KeyPair::KeyName

  CustomHTML:
    Type: String
    Default: "<h1>Hello from CloudFormation</h1>"

  LinuxAMI:
    Type: String
    Default: ami-02dfbd4ff395f2a1b

  WindowsAMI:
    Type: String
    Default: ami-01a15dfc48279bf55

# ---------------- CONDITIONS ----------------
Conditions:
  IsLinuxOnly: !Equals [!Ref OSSelection, Linux]
  IsWindowsOnly: !Equals [!Ref OSSelection, Windows]
  IsBoth: !Equals [!Ref OSSelection, Both]

  LaunchLinux: !Or
    - !Condition IsLinuxOnly
    - !Condition IsBoth

  LaunchWindows: !Or
    - !Condition IsWindowsOnly
    - !Condition IsBoth

# ---------------- NETWORK ----------------
Resources:

  VPC:
    Type: AWS::EC2::VPC
    Properties:
      CidrBlock: 10.0.0.0/16
      EnableDnsSupport: true
      EnableDnsHostnames: true
```

```yaml
IGW:
  Type: AWS::EC2::InternetGateway

AttachIGW:
  Type: AWS::EC2::VPCGatewayAttachment
  Properties:
    VpcId: !Ref VPC
    InternetGatewayId: !Ref IGW

Subnet:
  Type: AWS::EC2::Subnet
  Properties:
    VpcId: !Ref VPC
    CidrBlock: 10.0.1.0/24
    MapPublicIpOnLaunch: true
    AvailabilityZone: !Select [0, !GetAZs ""]

RouteTable:
  Type: AWS::EC2::RouteTable
  Properties:
    VpcId: !Ref VPC

Route:
  Type: AWS::EC2::Route
  DependsOn: AttachIGW
  Properties:
    RouteTableId: !Ref RouteTable
    DestinationCidrBlock: 0.0.0.0/0
    GatewayId: !Ref IGW

RouteAssoc:
  Type: AWS::EC2::SubnetRouteTableAssociation
  Properties:
    SubnetId: !Ref Subnet
    RouteTableId: !Ref RouteTable

SG:
  Type: AWS::EC2::SecurityGroup
  Properties:
    GroupDescription: Allow HTTP SSH RDP
    VpcId: !Ref VPC
    SecurityGroupIngress:
      - { IpProtocol: tcp, FromPort: 80, ToPort: 80, CidrIp: 0.0.0.0/0 }
```

```
      - { IpProtocol: tcp, FromPort: 22, ToPort: 22, CidrIp: 0.0.0.0/0 }
      - { IpProtocol: tcp, FromPort: 3389, ToPort: 3389, CidrIp: 0.0.0.0/0 }

# ---------------- LINUX ----------------
  LinuxEC2:
    Type: AWS::EC2::Instance
    Condition: LaunchLinux
    Properties:
      InstanceType: !Ref InstanceType
      KeyName: !Ref KeyName
      ImageId: !Ref LinuxAMI
      SubnetId: !Ref Subnet
      SecurityGroupIds: [!Ref SG]
      UserData:
        Fn::Base64: !Sub |
          #!/bin/bash
          yum install -y httpd
          systemctl start httpd
          systemctl enable httpd
          echo "<html><body><h2>LINUX SERVER</h2>${CustomHTML}</body></html>" >
/var/www/html/index.html

# ---------------- WINDOWS ----------------
  WindowsEC2:
    Type: AWS::EC2::Instance
    Condition: LaunchWindows
    Properties:
      InstanceType: !Ref InstanceType
      KeyName: !Ref KeyName
      ImageId: !Ref WindowsAMI
      SubnetId: !Ref Subnet
      SecurityGroupIds: [!Ref SG]
      UserData:
        Fn::Base64: !Sub |
          <powershell>
          Install-WindowsFeature -Name Web-Server
          Set-Content -Path "C:\inetpub\wwwroot\index.html" -Value
"<html><body><h2>WINDOWS SERVER</h2>${CustomHTML}</body></html>"
          Start-Service W3SVC
          </powershell>

# ---------------- OUTPUTS ----------------
Outputs:
  LinuxIP:
```

```
    Condition: LaunchLinux
    Value: !GetAtt LinuxEC2.PublicIp

  WindowsIP:
    Condition: LaunchWindows
    Value: !GetAtt WindowsEC2.PublicIp
```